

Session 07 Course Title: Mathematics II Course Code: MAT102/0541-102 Credits: 3	Course Teacher: Md. Yousuf Ali (YA) Lecturer, Department of GED. Email: yousufkumath@gmail.com Mobile: 01766906098
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(REF, RREF, NF)

Zero Row: All elements of the row are zero.

Non-Zero Row: At least one element of the row is non-zero.

Pivot: First non-zero element of each row considered from the left.

Example:

$$\begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 4 & 1/2 \\ -4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

ELEMENTARY ROW OPERATION:

- The interchange of the i th row and j th row; and it is denoted by $R_i \leftrightarrow R_j$.

Example:

$$M = \begin{pmatrix} 1 & 3 & 4 \\ 0 & 0 & 0 \\ 2 & 4 & 8 \end{pmatrix} \sim \begin{pmatrix} 1 & 3 & 4 \\ 2 & 4 & 8 \\ 0 & 0 & 0 \end{pmatrix} R_2 \leftrightarrow R_3$$

- The multiplication of every element of the i th row by a nonzero scalar k ; and it is denoted by $R_i \rightarrow kR_i$.

Example:

$$M = \begin{pmatrix} 1 & 3 & 4 \\ 0 & 0 & 0 \\ 2 & 4 & 8 \end{pmatrix} \sim \begin{pmatrix} 5 & 15 & 20 \\ 0 & 0 & 0 \\ 2 & 4 & 8 \end{pmatrix} R_1 \rightarrow 5R_1$$

- The addition of the elements of the i th row and k times the j th row; and it is denoted by $R_i \rightarrow R_i + kR_j$.

$$M = \begin{pmatrix} 1 & 3 & 4 \\ 0 & 0 & 0 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & 3 & 4 \\ 0 & 0 & 0 \\ 5 & 13 & 20 \end{pmatrix} \quad R_3 \rightarrow R_3 + 3R_1$$

Row Echelon Form (REF):

Step 1: All zero rows, if any, are at the bottom of the matrix.

Step 2: Each pivot in a row is to the right of the pivot in the preceding row.

N.B: All entries in a column below a pivot are zero.

$$\begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 4 & 1/2 \\ 0 & 0 & -4 & -4 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

REF

Procedure:

1. Interchange positions of zero rows (whenever they appear) so that they lie below all nonzero rows.
2. Selection of pivots.
3. Formation of zeroes below pivots (using pivots).

Example:

$$\begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 4 & 1/2 \\ 0 & 0 & 0 & 0 \\ -4 & 0 & 0 & 0 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 4 & 1/2 \\ -4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad R_4 \leftrightarrow R_3$$

$$\sim \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 4 & 1/2 \\ 0 & 8 & -4 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad R_3 \rightarrow R_3 + 4R_1$$

$$\sim \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 4 & 1/2 \\ 0 & 0 & -4 & -4 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad R_3 \rightarrow 3R_3 - 8R_2$$

REF

$R_3:$	-4	0	0	0
$4R_1:$	4	8	-4	0
$R_3 + 4R_1:$	0	8	-4	0

$3R_3:$	0	24	-12	0
$8R_2:$	0	24	32	4
$3R_3 - 8R_2:$	0	0	-44	-4

Reduced Row Echelon Form (RREF):**Step 1:** All zero rows, if any, are at the bottom of the matrix.**Step 2:** Each pivot in a row is to the right of the pivot in the preceding row.**Step 3:** Each pivot is equal to 1.**Step 4:** Each pivot is the only nonzero entry in its column.

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1/22 \\ 0 & 0 & 1 & 1/11 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

RREF**Procedure:**

1. Make each pivot as 1.
2. Make zeroes above each pivot by means of that pivot.

Example:

$$\begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 4 & 1/2 \\ 0 & 0 & 0 & 0 \\ -4 & 0 & 0 & 0 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 4 & 1/2 \\ 0 & 0 & -44 & -4 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \text{REF}$$

$$\sim \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 4 & 1/2 \\ 0 & 0 & 1 & 1/11 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad R_3 \rightarrow \frac{R_3}{-44}$$

$$\sim \begin{pmatrix} 1 & 2 & 0 & 1/11 \\ 0 & 3 & 0 & 3/22 \\ 0 & 0 & 1 & 1/11 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{matrix} R_1 \rightarrow R_1 + R_3 \\ R_2 \rightarrow R_2 - 4R_3 \end{matrix}$$

$$\sim \begin{pmatrix} 1 & 2 & 0 & 1/11 \\ 0 & 3 & 0 & 3/22 \\ 0 & 0 & 1 & 1/11 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{array}{l} R_1 \rightarrow R_1 + R_3 \\ R_2 \rightarrow R_2 - 4R_3 \end{array}$$

$$\sim \begin{pmatrix} 1 & 2 & 0 & 1/11 \\ 0 & 1 & 0 & 1/22 \\ 0 & 0 & 1 & 1/11 \\ 0 & 0 & 0 & 0 \end{pmatrix} R_2 \rightarrow \frac{R_2}{3}$$

$$\sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1/22 \\ 0 & 0 & 1 & 1/11 \\ 0 & 0 & 0 & 0 \end{pmatrix} R_1 \rightarrow R_1 - 2R_2$$

RREF

Normal Form (NF):

1. Consist of only ones and zeros.
2. Every row has a maximum of single one and rest are all zeros (there can be rows with all zeros).
3. Every column has a maximum of single one and rest are all zeros (there can be columns with all zeros).

Example:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 & | & 0 \\ 0 & 1 & 0 & | & 0 \\ 0 & 0 & 1 & | & 0 \end{pmatrix}$$

(I_3) $(I_3 \mid 0)$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \hline 0 & 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 & | & 0 \\ 0 & 1 & 0 & | & 0 \\ 0 & 0 & 1 & | & 0 \\ \hline 0 & 0 & 0 & | & 0 \end{pmatrix}$$

$(I_3 \mid 0)$ $(I_3 \mid 0)$

Normal Form of a matrix

$$(I_n)$$

$$(I_n \mid 0)$$

$$\begin{pmatrix} I_n \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} I_n & 0 \\ 0 & 0 \end{pmatrix}$$

To find Normal Form of a matrix, we need to do column operation,

$$\begin{array}{l}
 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1/22 \\ 0 & 0 & 1 & 1/11 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \boxed{\text{RREF}} \\
 \sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1/11 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad C_4 \rightarrow C_4 - \frac{1}{22}C_2 \\
 \sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad C_4 \rightarrow C_4 - \frac{1}{11}C_3
 \end{array}
 \quad \Bigg| \quad
 \begin{array}{l}
 \sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\
 \sim \begin{pmatrix} I_3 & 0 \\ 0 & 0 \end{pmatrix} \quad \boxed{\text{NF}}
 \end{array}$$